ROBOTICS

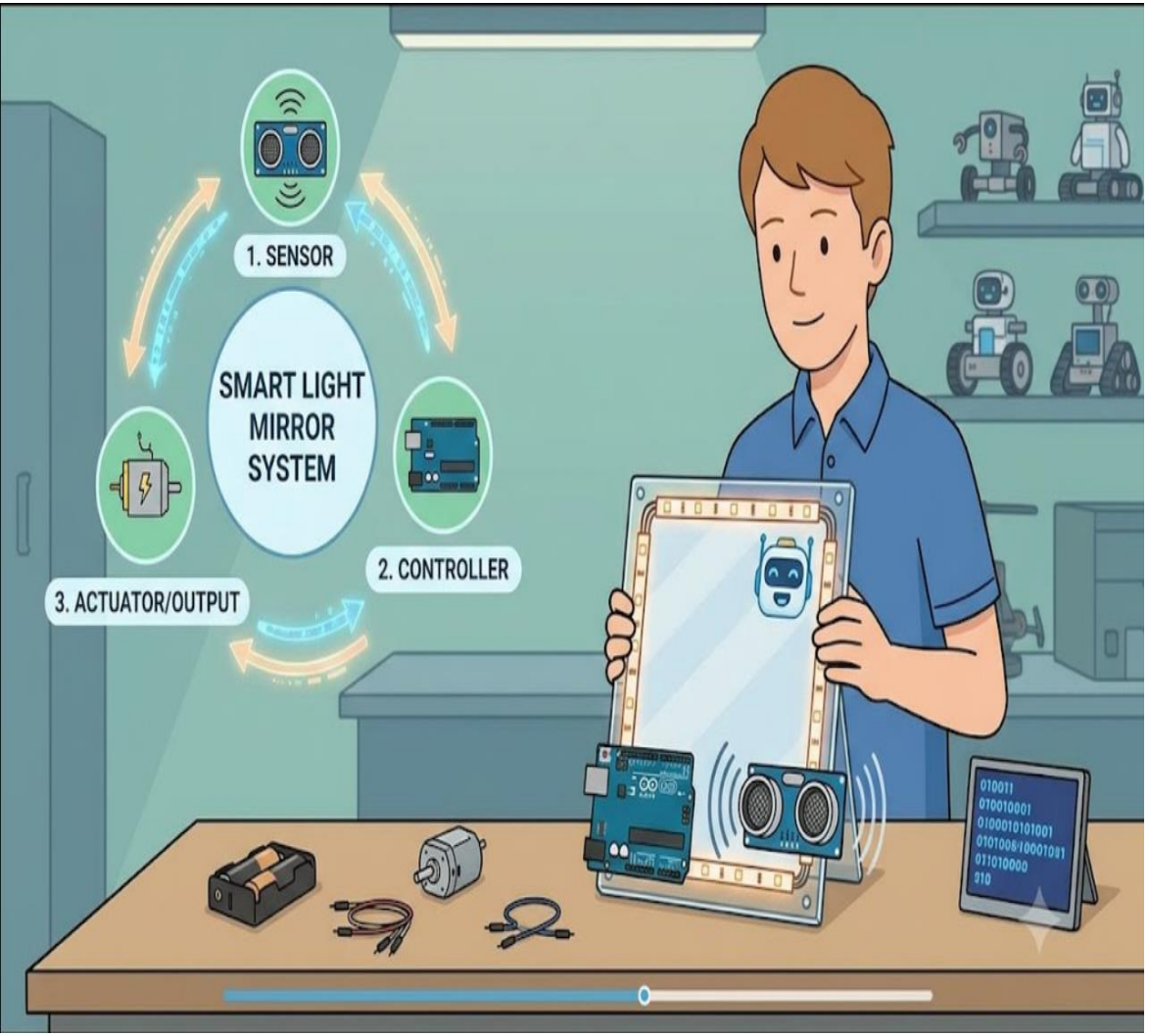
TECHNOLOGY AUTOMATION INNOVATION



ROBOTICS

Under the Supervisor
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Subject:DTIE-Projects

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Introduction to Robotics

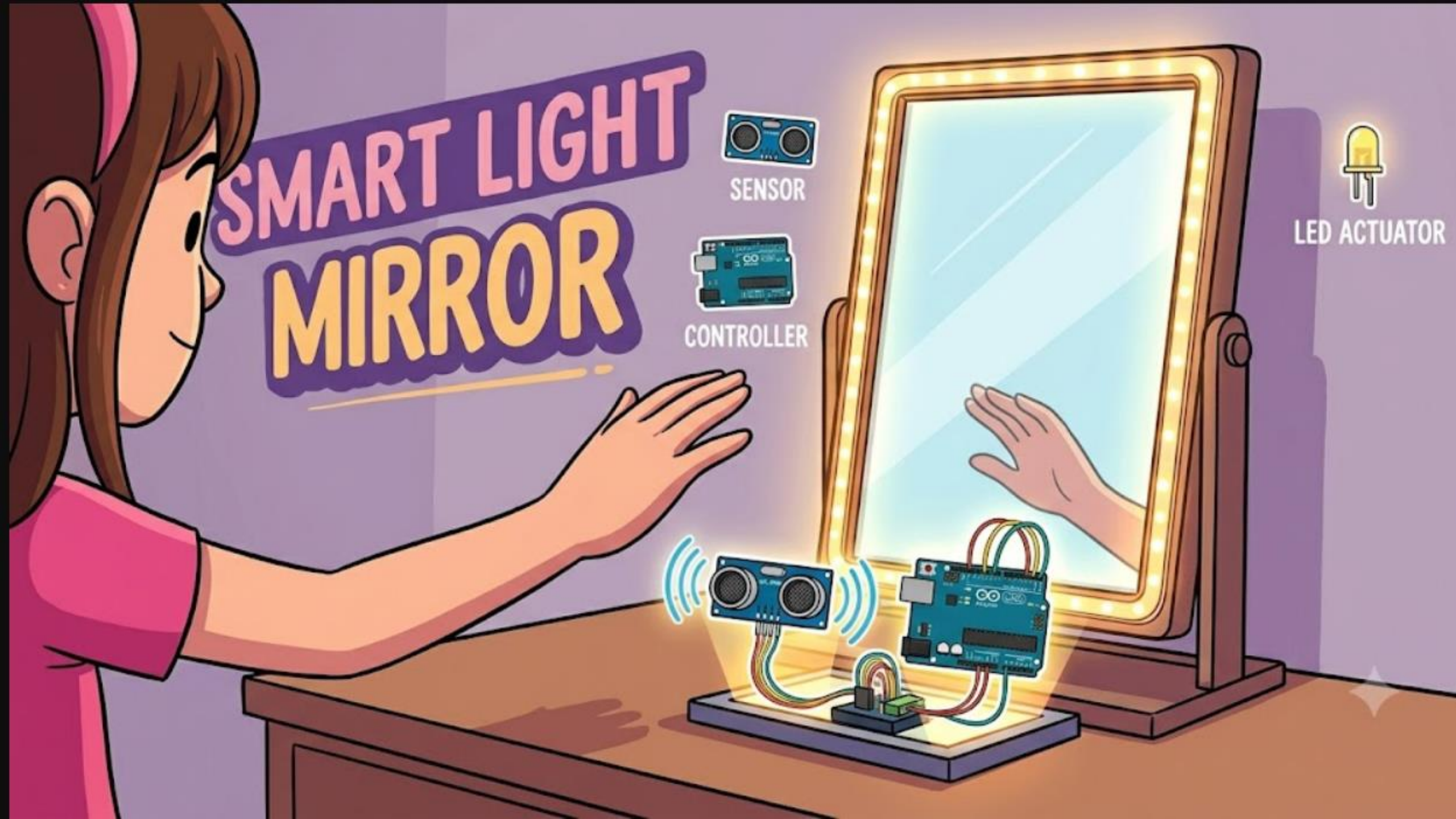
- **What is Robotics?**

- Robotics is a field that combines **mechanical systems, electronics, sensors and programming** to create systems that can sense their surroundings and perform actions automatically.

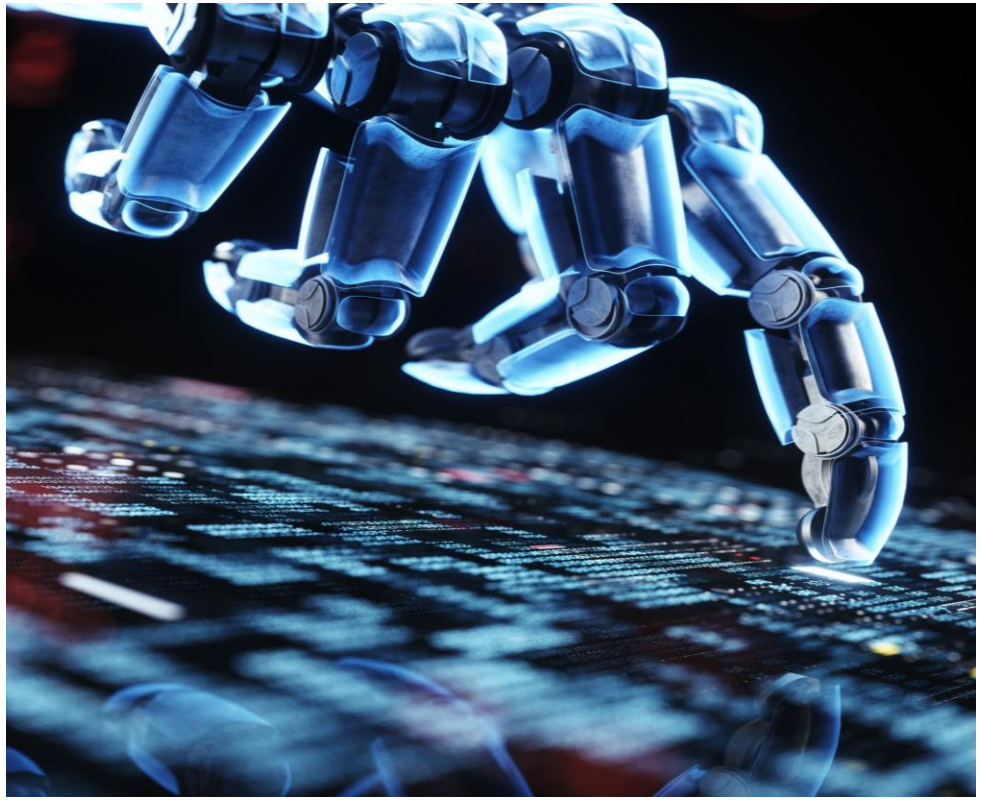
- **Key Elements of Robotics**
  - **Sensors** – detect changes in the environment
  - **Controller** – processes the sensor input
  - **Actuator/Output** – performs an action
  - **Programming** – controls the system's behavior
  - **Power Supply** – provides energy to the system

- **Our Project**

•Our Smart Light Mirror applies these principles using an **ultrasonic sensor, Arduino Uno and LED lighting**.


SMART LIGHT MIRROR
SENSOR
CONTROLLER
LED ACTUATOR

🤖 💡 Robotics Application: Smart Light Mirror



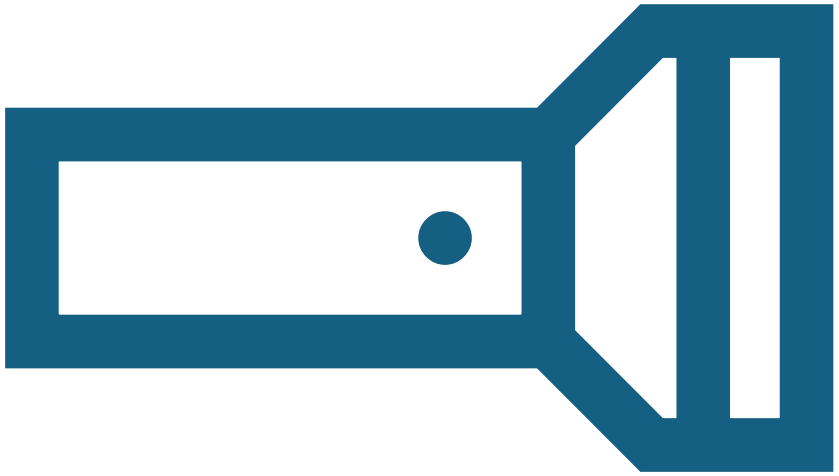
How Robotics is Applied
Our project follows the basic robotics cycle:
SENSE → PROCESS → ACT

In Our System

HC-SR04 Ultrasonic Sensor
↓
Detects the distance/presence of a person
Arduino Uno
↓
Processes the sensor reading using C++
LED Lights
↓
Automatically turn ON/OFF

Project Concept
The system automatically controls mirror lighting according to the user's proximity to the mirror.

Problem



- Conventional mirrors require manual control of lighting.
- Poor lighting can make mirror usage difficult.
- Users may need to reach for a switch every time.
- Lights can be left ON unnecessarily.
- A simple automatic lighting system can improve convenience.

Problem Statement

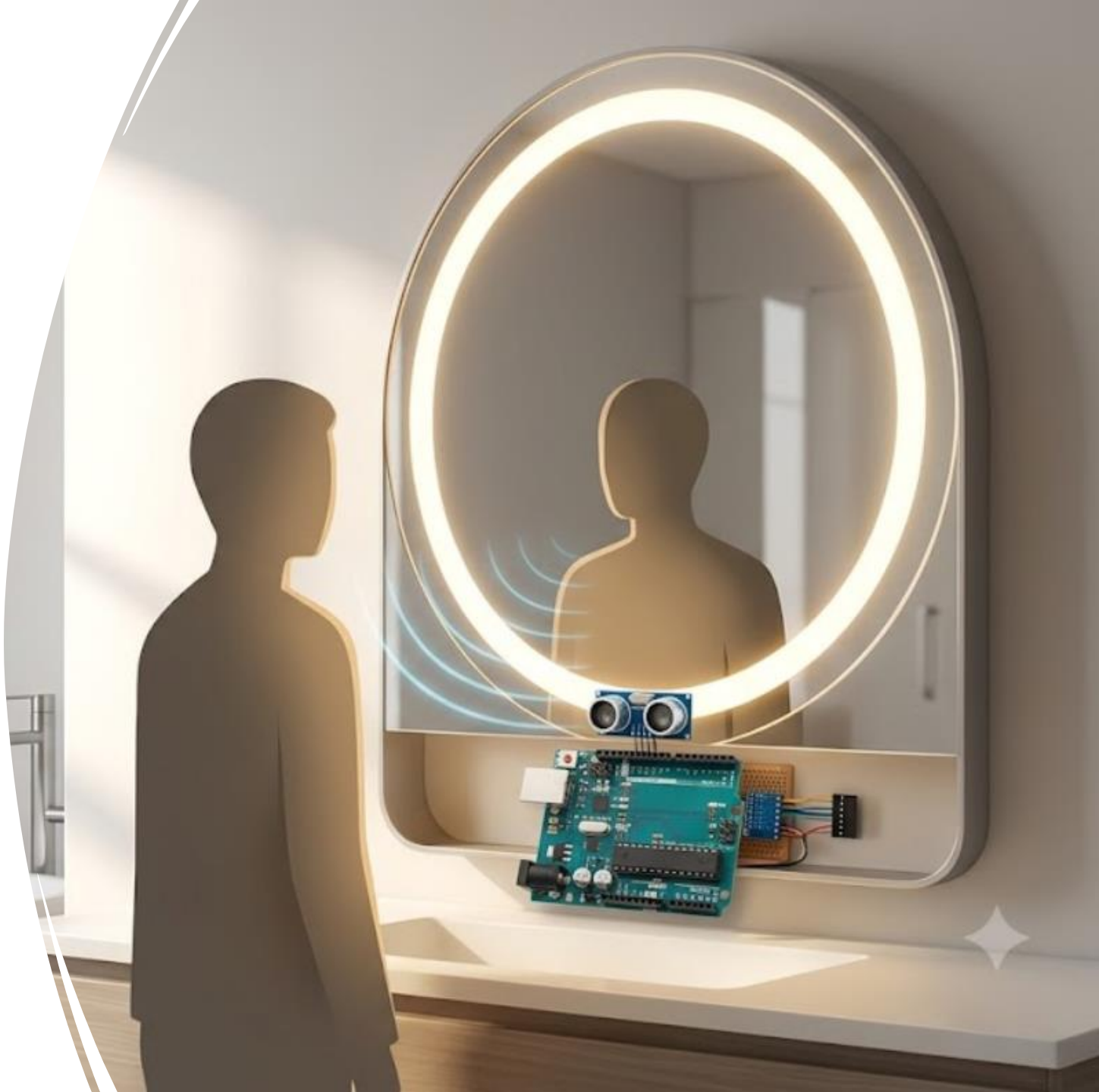
- **“How can we automatically control mirror lighting when a person approaches the mirror?”**

🧠 Problem: Need for Automation

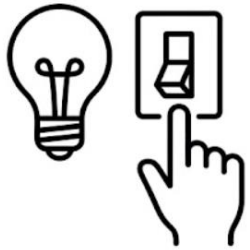
- **Problems with Conventional Mirror Lighting**
- Manual ON/OFF operation
- Inconvenient during daily use
- Unnecessary power consumption
- No automatic response to the user's proximity

- **Our Need**
- A system that can:
- **Detect Person → Process Distance → Control Light**

- This led us to develop an **Arduino-based Smart Light Mirror**

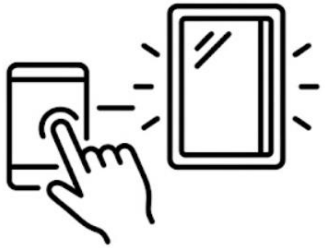


Empathy









-
- **Target Users**
  - Students
  - Home users
  - Working professionals
  - People using mirrors in low-light conditions

- **User Difficulties**
  - Need to manually switch the light
  - Poor visibility without proper illumination
  - Forgetting to switch lights OFF
  - Want a convenient and automatic system

Empathy Map

USER SAYS

- “I want the mirror light to turn ON automatically.”
- **USER THINKS**
- “Why should I manually operate the light every time?”

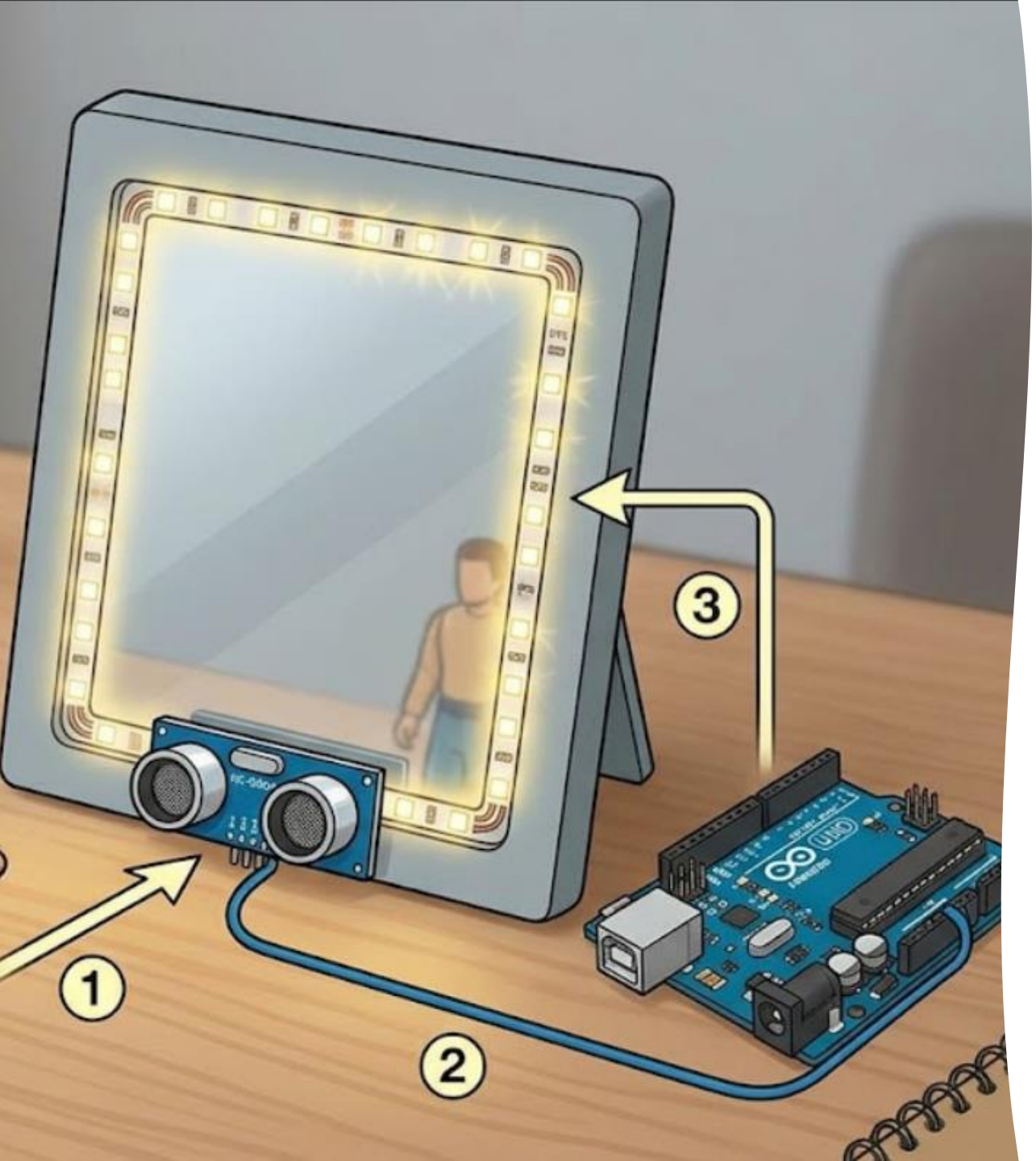
USER FEELS

- Inconvenienced
- Wants easier operation
- Wants better lighting

USER NEEDS

- Automatic lighting
- Quick response
- Simple operation
- Efficient use of electricity





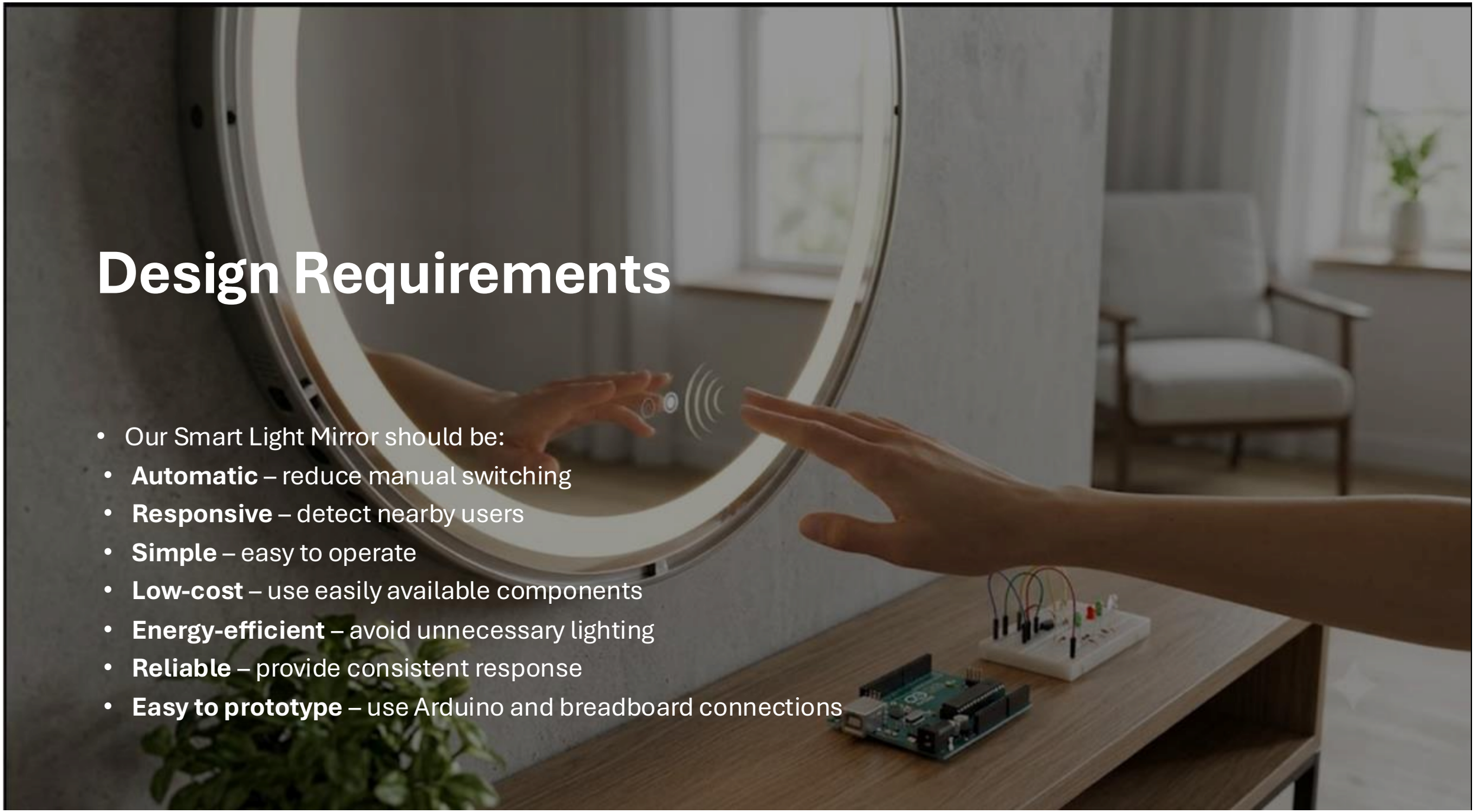
Define

- **Defined Problem**
- Users need convenient mirror lighting that automatically responds when they approach the mirror.

- **Defined Solution:**
- Develop a **Smart Light Mirror** using:
- **HC-SR04 + Arduino Uno + LED Lights**

- **Main Objective:**
- To create a simple system that:
- Detects the user's proximity
- Processes the distance
- Automatically controls the mirror lighting

Design Requirements



- Our Smart Light Mirror should be:
  - **Automatic** – reduce manual switching
  - **Responsive** – detect nearby users
  - **Simple** – easy to operate
  - **Low-cost** – use easily available components
  - **Energy-efficient** – avoid unnecessary lighting
  - **Reliable** – provide consistent response
  - **Easy to prototype** – use Arduino and breadboard connections



Idea Generation

- We considered different methods for automatic mirror lighting:
  - Manual switch-controlled mirror
  - IR sensor-based lighting
  - LDR-based lighting
  - PIR sensor-based lighting
  - Ultrasonic sensor-based lighting
  - Touch-controlled lighting
  - HC-SR04-based Smart Light Mirror

- **Selected Idea**
- We selected an **ultrasonic sensor-based system** because it can measure the distance between the person and the mirror without physical contact.

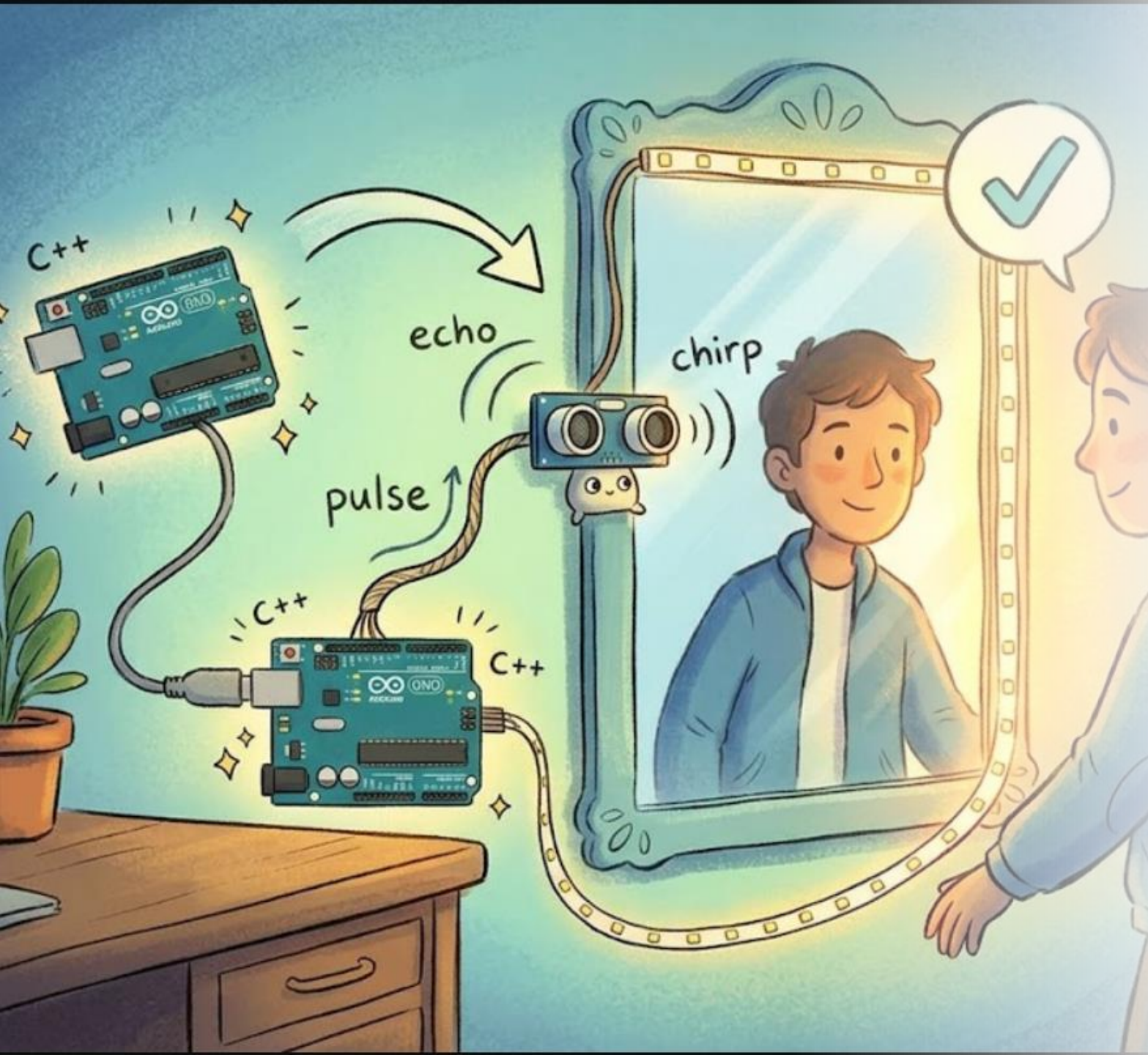
5. Multi Voting

We compared the ideas using:

- Cost
- Detection capability
- Ease of programming
- Availability of components
- Response
- Prototype feasibility

Method	Cost	Detection	Programming	Overall
Manual Switch	High	Low	Easy	⭐⭐⭐
IR Sensor	Low	Medium	Easy	⭐⭐⭐⭐
LDR	Low	Light only	Easy	⭐⭐⭐
PIR	Low	Good	Easy	⭐⭐⭐⭐
HC-SR04	**Low**	**Distance-based**	**Easy**	⭐⭐⭐⭐⭐

Final Selection
HC-SR04 + Arduino Uno



Rapid Estimation & Solution

- **Proposed Solution**
- The **HC-SR04 ultrasonic sensor** measures the distance between the person and the mirror.
- The Arduino Uno reads this distance and controls the LEDs according to the programmed condition.

- **System Flow**
- Person Approaches

↓

HC-SR04 Sends Ultrasonic Signal

↓

Echo is Received

↓

Arduino Calculates Distance

↓

Distance Condition is Checked

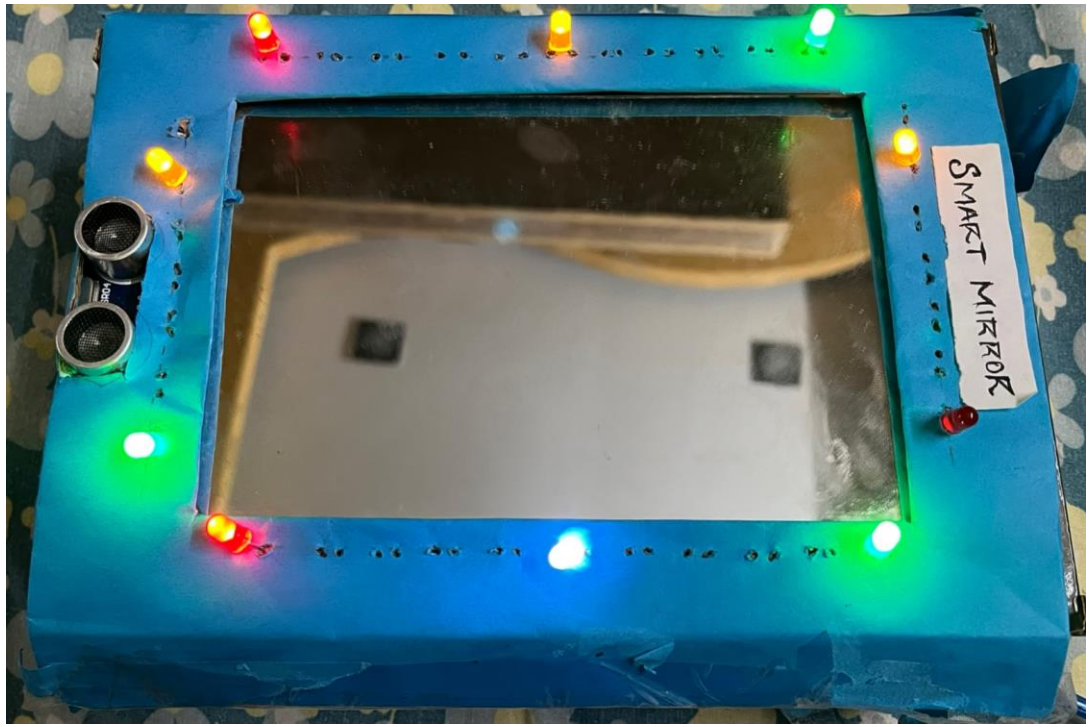
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LEDs Turn ON/OFF

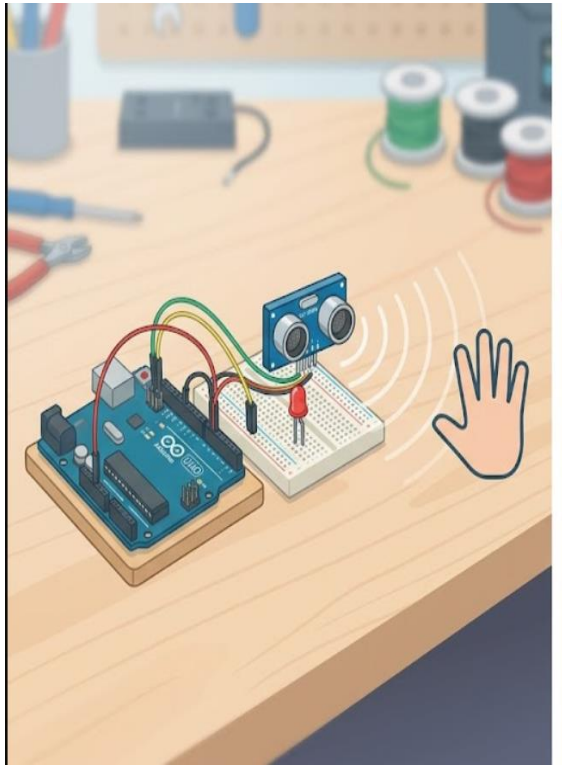
- **Programming**
- The system is programmed in **C++ using Arduino IDE**.

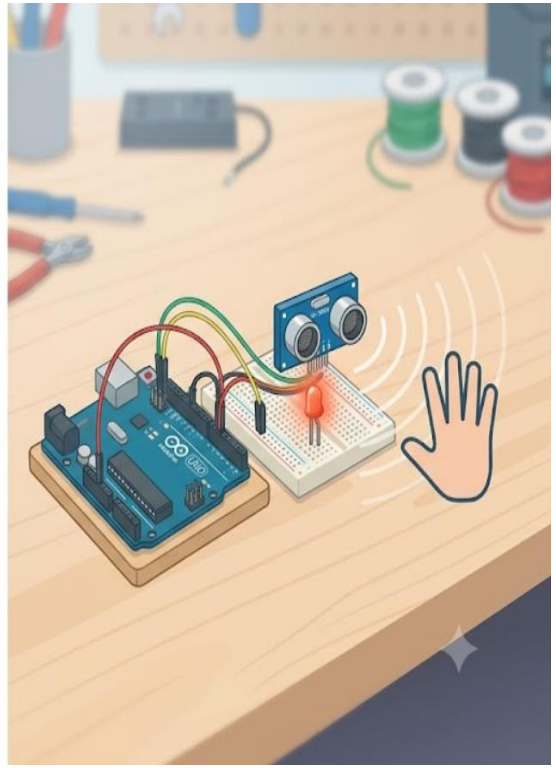
7. Rough Prototype

- **Components Used**
  - Arduino Uno
  - HC-SR04 Ultrasonic Sensor
  - LED Lights
  - Breadboard
  - Jumper Wires
  - Mirror
  - Power/USB connection



Rough Prototype





Circuit Connection

HC-SR04

- VCC → Arduino 5V
- GND → Arduino GND
- TRIG → Arduino digital pin
- ECHO → Arduino digital pin

LED

- Connected to Arduino output through the breadboard
- Jumper wires are used for connections

Testing

Distance Condition	Expected Action
Person far away	LED OFF
Person enters detection range	LED ON
Person remains nearby	LED remains ON
Person moves away	LED OFF

Software

The Arduino is programmed through the **Arduino IDE using C++**.

Demonstration

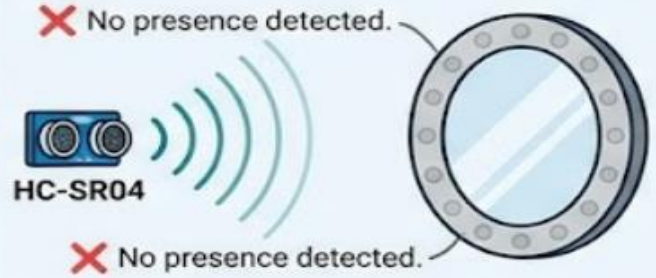
SMART MIRROR PROTOTYPE DEMONSTRATION

1 SYSTEM START-UP



Arduino Uno connected. System **ON**.

2 IDLE STATE - EMPTY RANGE



LEDs remain **OFF**.

3 OBJECT APPROACHES



User approaches. Distance: Decreasing.

4 PROCESSING DETECTION



Distance processed. LEDs turn **ON!**

5 OBJECT RETREATS



User leaves. Distance: Increasing.

6 IDLE STATE - RESET



No presence. LEDs turn **OFF**.

DEMONSTRATION RESULT

The prototype demonstrates automatic mirror lighting using ultrasonic distance sensing and Arduino-based control.




THANK
YOU